

Astrophysical implications of new light Higgs bosons

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We explore the consequences of the existence of a new light pseudoscalar particle, such as the axion. Thermal energy can be converted into axions through annihilation ($e^+e^- \rightarrow \gamma h$), photoproduction ($\gamma e \rightarrow he$), and plasma decay ($\gamma_p \rightarrow \gamma h$). Explicit expressions for the cross sections and decay lifetimes are given. We conclude that axion production can occur in advanced stages of stellar evolution and in the big-bang creation of the universe. The most dramatic effect, however, is expected during gravitational collapse of massive stars where axions may help trigger a supernova explosion.

It has been recently pointed out by Weinberg¹ and Wilczek² that a light pseudoscalar Higgs boson, called an axion, is a necessary consequence of theories³ which unify the strong, weak, and electromagnetic interactions and which possess a global U(1) chiral symmetry to avoid naturally CP violation in strong interactions.⁴ Though reactor and beam-dump experiments argue against such a particle,^{1,5} so far the analysis of these experiments and of the nuclear reactions involving axions is too crude to definitely rule out their existence.⁶ New experiments to search for axions have been proposed.^{7,8}

Here we shall concentrate on the astrophysical implications of such light pseudoscalar mesons. In principle, axions could be created in three different astrophysical situations: during the big-bang birth of the universe, during nuclear burning in stellar evolution, and in the final gravitational collapse of massive stars. We shall argue that within the range of parameters predicted for axions, they play a most interesting role in gravitational collapse.

In this context one naturally compares axions with photons and neutrinos: The semiweak interaction between matter and axions implies that $\sigma_\gamma \gg \sigma_h \gg \sigma_\nu$, where σ_γ , σ_h , and σ_ν are typical

production and absorption cross sections for photons, axions, and neutrinos (we shall be more precise below). Axions have thus the potential to act *both* as a mechanism for energy transport *and* energy loss. Which role they play depends on the stellar structure of the axion-emitting region.

We shall assume, for orientation, that $50 \text{ keV}/c^2 \leq m_h \leq 200 \text{ keV}/c^2$, and that the axion decays through $h \rightarrow \gamma\gamma$ with $0.01 \text{ sec} \leq \tau_h \leq 1 \text{ sec}$. The threshold temperature is $T_{\text{threshold}} = (m_h c^2)/k \approx (1-4) \times 10^9 \text{ K}$. We shall consider the production of axions in pure leptonic processes for which exact expressions can be derived. The interaction Lagrangian is¹

$$\mathcal{L}_{\text{int}} = g \bar{\psi}_e \gamma_5 \psi_e h = 2^{1/4} G_F^{1/2} m_e c_e \bar{\psi}_e \gamma_5 \psi_e h, \quad (1)$$

where $c_e = \tan \alpha$ or $\cot \alpha$ and is of order one. We will set $c_e = 1$ in numerical calculations.

We shall consider only three processes, shown in Fig. 1, for axion creation: (i) annihilation, (ii) photoproduction, and (iii) plasma decay.

(i) For electron-positron annihilation, Fig. 1(a),

$$e^+ + e^- \rightarrow \gamma + h, \quad (2)$$

we find the following total cross section:

$$\sigma(e^+e^- \rightarrow \gamma h) = \frac{\alpha g^2}{2s\beta^2} \left[\left(1 + m_h^2/s - 2(m_h^2/s)(2m_e^2 - m_h^2)/(s - m_h^2) \right) \ln \left(\frac{1+\beta}{1-\beta} \right) - 2m_h^2\beta/(s - m_h^2) \right], \quad (3)$$

where $\beta = (1 - 4m_e^2/s)^{1/2}$ and s is the square of the c.m. energy. This cross section is shown in Fig. 2 for $m_h = 50 \text{ keV}/c^2$. It depends very weakly on m_h . Assuming, as we have done, that $m_h < 2m_e$, we find that $\sigma(e^+e^- \rightarrow \gamma h)$ diverges like $1/\beta$ as $\beta \rightarrow 0$ near threshold where e^+e^- annihilate from rest.⁹

(ii) Photoproduction of axions,

$$\gamma + e \rightarrow h + e, \quad (4)$$

is shown in Fig. 1(b). We find

$$\sigma(\gamma e \rightarrow he) = \frac{\alpha g^2}{8s} \left(\frac{p}{k} \right) \left[-3 + m_e^2/s + 3m_h^2/(s - m_e^2) + m_e^2 m_h^2/s(s - m_e^2) + 4m_h^2(s + m_e^2)/(s - m_e^2)^2 \right. \\ \left. + \left(\frac{\sqrt{s}}{p} \right) \left[1 - 2m_h^2/(s - m_e^2) - 2m_h^2(2m_e^2 - m_h^2)/(s - m_e^2)^2 \right] \ln \left(\frac{p_0 k_0 + pk - m_h^2}{p_0 k_0 - pk - m_h^2} \right) \right], \quad (5)$$

where

$$p_0 = (s - m_e^2 + m_h^2)/2\sqrt{s}, \quad p = (p_0^2 - m_h^2)^{1/2}, \quad k_0 = (s + m_e^2)/2\sqrt{s}, \quad \text{and } k = \sqrt{s} - k_0.$$

In Fig. 2 we have plotted $\sigma(\gamma e \rightarrow he)$ as a function of $s/4m_e^2$ for $m_h = 50 \text{ keV}/c^2$. Except near threshold, this cross section is not sensitive to the value of m_h .

(iii) Plasma decay,

$$\text{plasmon } (k) \rightarrow \gamma(k') + h, \quad (6)$$

as shown in Fig. 1(c) is somewhat more difficult to calculate because of the loop integral. The final photon may also represent a plasma oscillation for which the four-vector square k'^2 does not vanish. The amplitude is proportional to the three-point function

$$A_{\mu\nu}(k, k') = -2 \int \frac{d^4 l}{(2\pi)^4} \text{Tr}[\gamma_\mu G(l) \gamma_\nu G(l - k') \gamma_5 G(l - k)], \quad (7)$$

where $G(l)$ is the Green's function for the propagation of an electron in the plasma.¹⁰ We have evaluated this function for arbitrary k setting $k'^2 = 0$ and keeping only the free electron propagator in the Green's functions appearing in Eq. (7), i.e., $G(l) \rightarrow 1/(\not{l} - m_e)$, etc. We find

$$A_{\mu\nu}(k, k') = \epsilon_{\mu\nu\alpha\beta} k^\alpha k'^\beta A(k, k'), \quad (8)$$

where A is in general a complex amplitude

$$A(k, k') = \frac{m_e}{\pi^2(k^2 - m_h^2)} [J(m_h^2/m_e^2) - J(k^2/m_e^2)] \quad (9)$$

$\substack{k'^2=0 \\ 2k \cdot k' = k^2 - m_h^2}$

with

$$J(x^2) = \left\{ \theta(1 - x/2) \cos^{-1}(x/2) + \theta(x/2 - 1) i \ln \left[x/2 + \left(\frac{x^2}{4} - 1 \right)^{1/2} \right] - \pi/2 \right\}^2. \quad (10)$$

As expected, the imaginary part of A comes from $J(k^2/m_e^2)$ for $k^2 > 4m_e^2$.

Both longitudinal and transverse plasmon decay are allowed, and we find that the decay widths are

$$\Gamma_l = \tau_l^{-1} = \frac{\alpha^2 g^2 m_e^2}{3\pi^3 k^2 (\partial \epsilon^l / \partial \omega)} (1 - m_h^2/k^2) |J(m_h^2/m_e^2) - J(k^2/m_e^2)|^2 \quad (11)$$

with a similar expression for $\Gamma_t = \tau_t^{-1}$ in which $k^2 \partial \epsilon^l / \partial \omega$ is replaced by $\partial(\omega^2 \epsilon^t) / \partial \omega$. Here $\omega = k_0$, and ϵ^l and ϵ^t are the longitudinal and transverse dielectric constants,¹⁰ respectively.

What astrophysical processes are likely to be

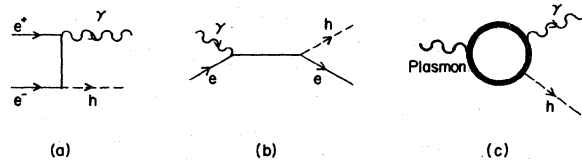


FIG. 1. Feynman diagrams for axion production through (a) annihilation (b) photoproduction, and (c) plasma decay. In each case there is an additional Feynman diagram obtained by interchanging the photon and axion lines.

affected by axion production?

(a) We do not expect much axion creation during normal stellar evolution because temperatures range between 10^7 to $10^9 \text{ }^\circ\text{K}$ for hydrogen through carbon burning. Clearly for photoproduction and plasma decay we need $T \gtrsim T_{\text{threshold}}$. Though e^+e^- pairs have enough mass energy to annihilate into axions, the number of such pairs is very small at low temperatures. In Fig. 3 we plot $R_h(T, \mu_e)$, the axion production rate per $\text{cm}^3 \text{ sec}$, defined by

$$R_h(T, \mu_e) = \int \int dn_{e^+} dn_{e^-} v \sigma(e^+ e^- \rightarrow \gamma h), \quad (12)$$

where $\vec{v}$ is the relative velocity between e^+ and e^- , and

$$dn_{e^\pm} = 2[1 + \exp(E_\pm/kT \pm \mu_e/kT)]^{-1} d^3 p_\pm / (2\pi)^3$$

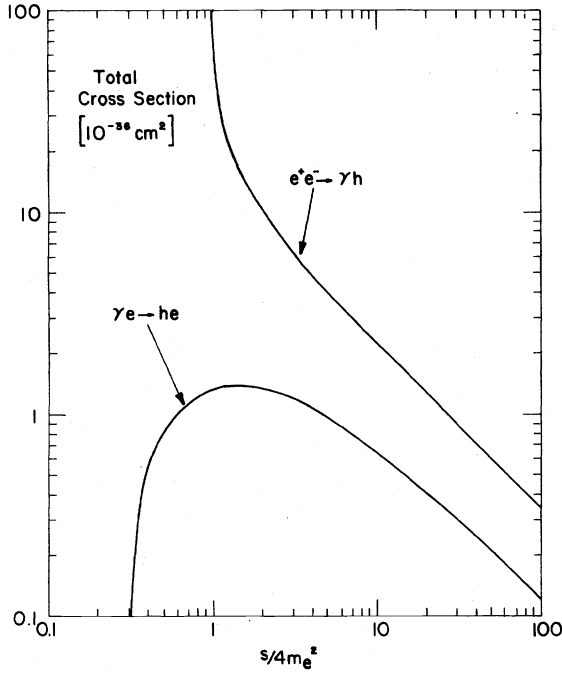


FIG. 2. Total cross section for $e^+e^- \rightarrow \gamma h$ and $\gamma e \rightarrow h e$ [Eqs. (3) and (5) in the text] for the case $m_h = 50 \text{ keV}/c^2$, in units of 10^{-36} cm^2 .

are the $e^\pm$ distribution functions for a given electron chemical potential μ_e . $R_h(T, \mu_e)$ is a very steep function of T . The ranges of T and μ_e shown in Fig. 3 describe the conditions during gravitational collapse which we shall treat separately. We expect, however, some axion production during the advanced stages of evolution in massive stars, namely oxygen through silicon burning when $T \approx (1.5-3.5) \times 10^9 \text{ K}$. Axion emission will speed up these processes because they transport energy much more readily than photons. We can find no observational consequences of this speed-up.

It is interesting to note that above a certain temperature T^* neutrino production dominates over axion production because $\sigma_\nu \sim s$ and $\sigma_h \sim 1/s$. For e^+e^- annihilation, $\sigma(e^+e^- \rightarrow \nu\bar{\nu}) \approx G_F^2 s$ and $\sigma(e^+e^- \rightarrow \gamma h) \approx \alpha m_e^2 G_F/s$, and we can estimate T^* by requiring these two cross sections to be equal:

$$T^* = \frac{1}{k} \sqrt{s^*} = \frac{1}{k} (\alpha m_e^2 / G_F)^{1/4} \approx 1.3 \times 10^{12} \text{ K}.$$

In Fig. 3 we have also plotted the neutrino pair production rate per $\text{cm}^3 \text{ sec}$,

$$R_\nu(T, \mu_e) = \int \int dn_{e^+} dn_{e^-} v \sigma(e^+e^- \rightarrow \nu\bar{\nu}), \quad (13)$$

and we have summed over the three kinds of

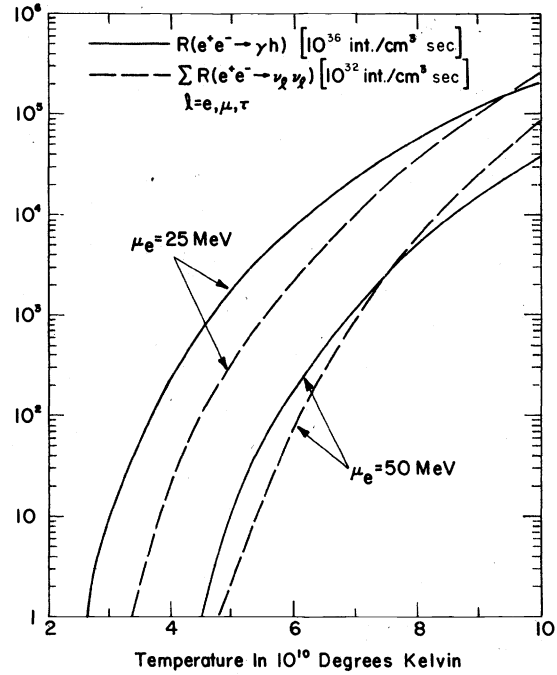


FIG. 3. Axion production rates per unit volume as a function of temperature for two values of the electron chemical potential μ_e , assuming $m_h = 100 \text{ keV}/c^2$. The dashed lines are the $\nu\bar{\nu}$ pair production rates per unit volume summed over ν_e, ν_μ , and ν_τ . The axion rates are given in units of 10^{36} interactions/ $\text{cm}^3 \text{ sec}$, and the $\nu\bar{\nu}$ rates in units of 10^{32} interactions/ $\text{cm}^3 \text{ sec}$.

neutrinos ν_e, ν_μ , and ν_τ .¹¹ Indeed we find that $R_\nu(T, \mu_e) > R_h(T, \mu_e)$ for $T \gtrsim 1.3 \times 10^{12} \text{ K}$.

As can be seen from Fig. 3, these production rates are strongly temperature dependent. Changing T by two orders of magnitude changes R_h by 20 and R_ν by 27 orders of magnitude. Table I illustrates this point for $\mu_e = 25 \text{ MeV}$. Note that at very high temperatures, $T \gtrsim 10^{12} \text{ K}$, the production rates become independent of the electron chemical potential μ_e .

We will discuss plasma decay and photoproduction only briefly. These reactions may be more important than the annihilation reaction particularly at low temperatures. Figure 3 and Table I may be used to estimate the ratio of axion versus neutrino production. In fact, we expect these

TABLE I. Axion and neutrino pair production rates per $\text{cm}^3 \text{ sec}$ as a function of temperature T : $T_{10} = T/10^{10} \text{ K}$, $\mu_e = 25 \text{ MeV}$, $m_h = 100 \text{ keV}/c^2$.

T_{10}	$R(e^+e^- \rightarrow \gamma h)$	$R(e^+e^- \rightarrow \nu\bar{\nu})$
1	2.7×10^{26}	2.4×10^{19}
10	2.0×10^{41}	2.5×10^{37}
125	1.7×10^{46}	2.1×10^{46}

ratios to be approximately equal, i.e.,

$$\frac{R(e^+e^- \rightarrow \gamma h)}{R(e^+e^- \rightarrow \nu\bar{\nu})} \approx \frac{R(\text{plasmon} \rightarrow \gamma h)}{R(\text{plasmon} \rightarrow \nu\bar{\nu})}. \quad (14)$$

This ratio is also temperature dependent, but its T dependence is relatively mild. From Table I, it changes by only seven orders of magnitude for T between 10^{10} K and 10^{12} K. Note, furthermore, that this ratio increases at lower temperatures where the plasma process becomes important.

The dominant reaction at still lower temperatures, e.g. $T \approx 10^9$ K, is photoproduction. We shall again relate the ratio of axion production to neutrino pair production and estimate that

$$\begin{aligned} \frac{R(\gamma e^- \rightarrow h e^-)}{R(\gamma e^- \rightarrow \nu\bar{\nu} e^-)} &\approx \frac{m_e^2}{G_F (kT)^4} \\ &\approx \frac{4.1 \times 10^{10}}{T_{10}^4}, \end{aligned} \quad (15)$$

where T_{10} is the temperature in units of 10^{10} K. Therefore, we find that only above $T^* \approx 4.5 \times 10^{12}$ K will neutrino pair production exceed axion production in all three reactions.

(b) Such high temperatures are presumably encountered only during the very early stages of the expansion of the universe, in the so-called hadron era about which not much is known. Assuming the standard big-bang theory, axions were present in large numbers and were as plentiful as photons with which they were in thermal equilibrium. The subsequent evolution of the axion gas depends on the interaction strength, mass, and lifetime of the axion. Given our earlier assumptions about these properties, we can say that the axions decayed sometime after the neutrinos decoupled from matter ($T \approx 10^{11}$ K, $t_{\text{expansion}} \approx 10^{-2}$ sec), but probably before e^+e^- annihilation stopped ($T \approx 10^9$ K, $t_{\text{expansion}} \approx 10^2$ sec). In this case the only observable effect would be a distortion of the present 2.7 K black-body radiation. However, since the decay occurred very early, the decay photons would have completely thermalized, that is, the fluctuation in the radiation field caused by axion decay in such early times is quickly washed out.¹²

(c) Axion production during gravitational collapse is a most promising mechanism for triggering supernova explosions. Occurring in the

temperature range $T \approx 10^{10} - 10^{11}$ K, well above threshold yet not high enough for the neutrinos to dominate, conversion of thermal energy into axions proceeds via all of the three reactions discussed above. In this temperature range which corresponds to $s/4m_e^2 \approx 0.7 - 70$, we have seen that these reactions have rather large cross sections, $10^{-35} - 10^{-36}$ cm², so the energy conversion proceeds rapidly. The deposition of this energy into the material around the imploding core can help blow off the envelope.¹³

The following scenario seems plausible: Much of the gravitational binding energy of the core is still carried away by neutrinos from neutronization; however, a few percent is turned into heat as the core collapses. This thermal energy is converted into $\nu\bar{\nu}$ pairs and into axions by e^+e^- annihilation, photoproduction, and plasma decay. The $\nu\bar{\nu}$ escape readily, within 0.1 to 1 sec, but the axions slowly diffuse out (our estimates range between 1 to 10 h). While the neutrinos pass through the envelope with very few interactions, the axions deposit almost all their energy in the extended envelope either by direct absorption or through their decay photons which of course are readily absorbed in the envelope. We estimate that about 10^{50} ergs can be transported by axions and deposited in the envelope.

This is a crude estimate for a very simple scenario, however, the advantages and disadvantages of the axion over the neutrino in triggering supernova explosions are clear: Neutrinos carry away energy promptly from the hot dense core, but deliver very little of it to the envelope. Axions, on the other hand, by decaying into photons deposit all their energy inside the envelope, but take a relatively long time to diffuse out of the core.

One can easily list other reactions for axion production, however, we doubt that these will change the basic picture emerging from our analysis: The existence of an axion would have a small effect on big-bang theories and on stellar evolution, but would have dramatic consequences during gravitational collapse.

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⁹The decay rate remains finite. For orthopositronium decay from the group state (1^3S_1) into γh , we find

$$\tau_{\text{ortho}} = \frac{24\pi}{\alpha^4 g^2 m_e (1 - m_h^2/4m_e^2)} \approx 8 \text{ sec.}$$

Parapositronium (1^1S_0) decay into γh is forbidden.

¹⁰See, e.g., V. N. Tsytovich, Zh. Exp. Teor. Fiz. **40**, 1775 (1961) [Sov. Phys.—JETP **13**, 1249 (1961)].

¹¹ ν_μ and ν_τ are created by weak neutral currents only, for which we have assumed e - μ - τ universality and used the standard Weinberg-Salam model with $\sin^2\theta_W = 0.3$.

¹²Unless, of course, the axion has a very long lifetime. One may then put limits on the mass and lifetime of the axion, as has been done for new neutral leptons. See B. Lee and S. Weinberg, Phys. Rev. Lett. **39**, 165 (1977); D. A. Dicus, E.N. Kolb, and V. Teplitz, *ibid.* **39**, 168 (1977).

¹³S. Colgate and R. White originally suggested [Astrophys. J. **143**, 626 (1966)] that a supernova explosion is triggered when neutrinos from the imploding core deposit their energy in the envelope. Of course the difficulty with this theory is that neutrinos interact very little in the envelope.